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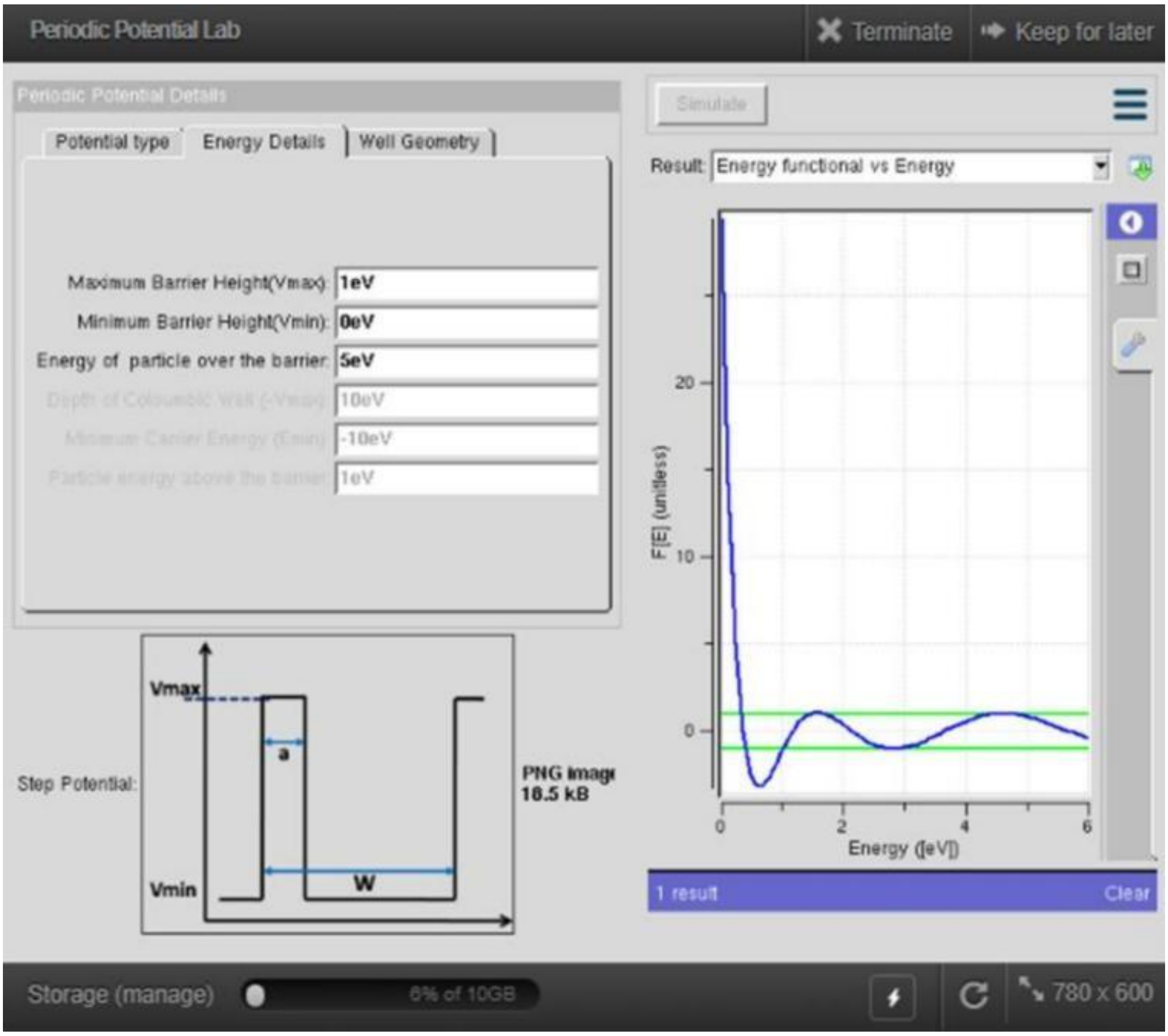
Exp. 4. Kronig Penney Model

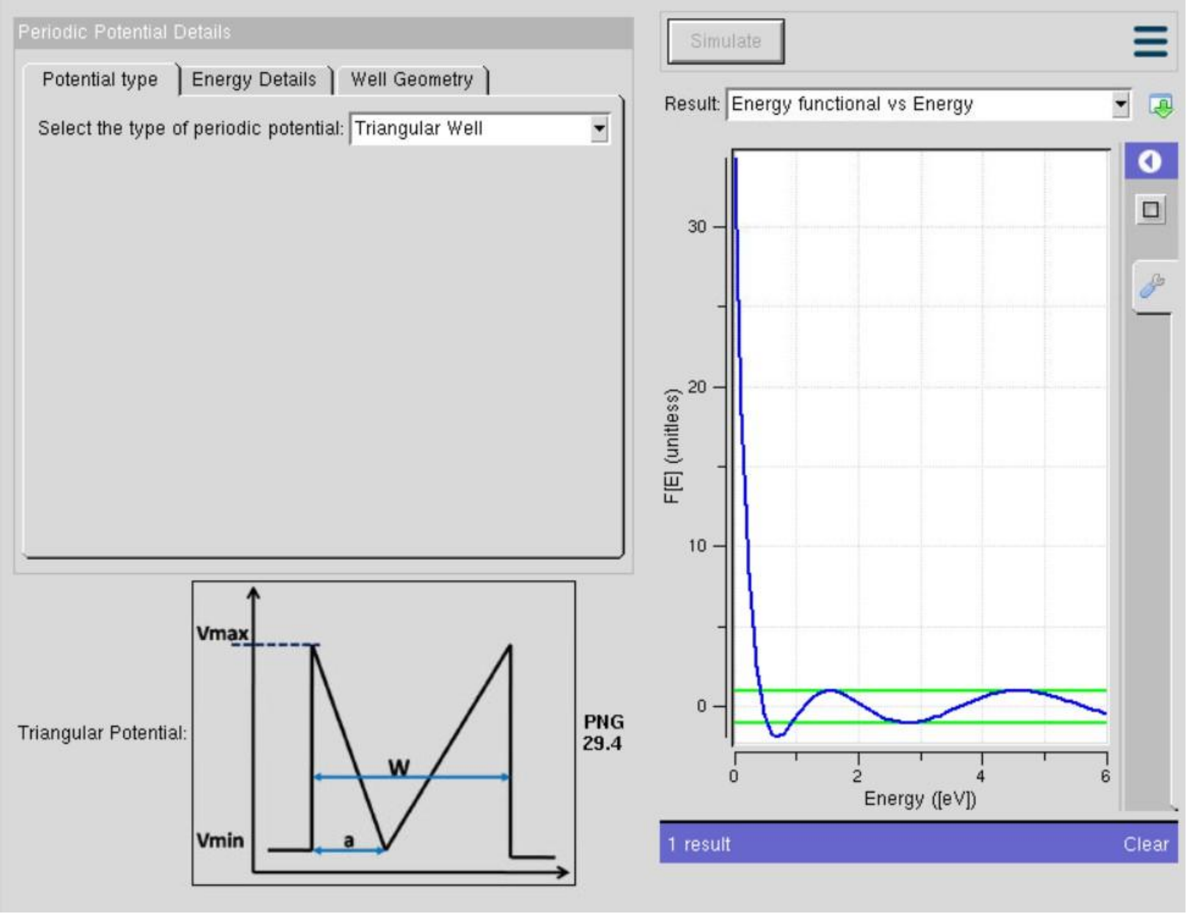
Aim:

To simulate Energy functional vs Energy graph using Kronig Penney model



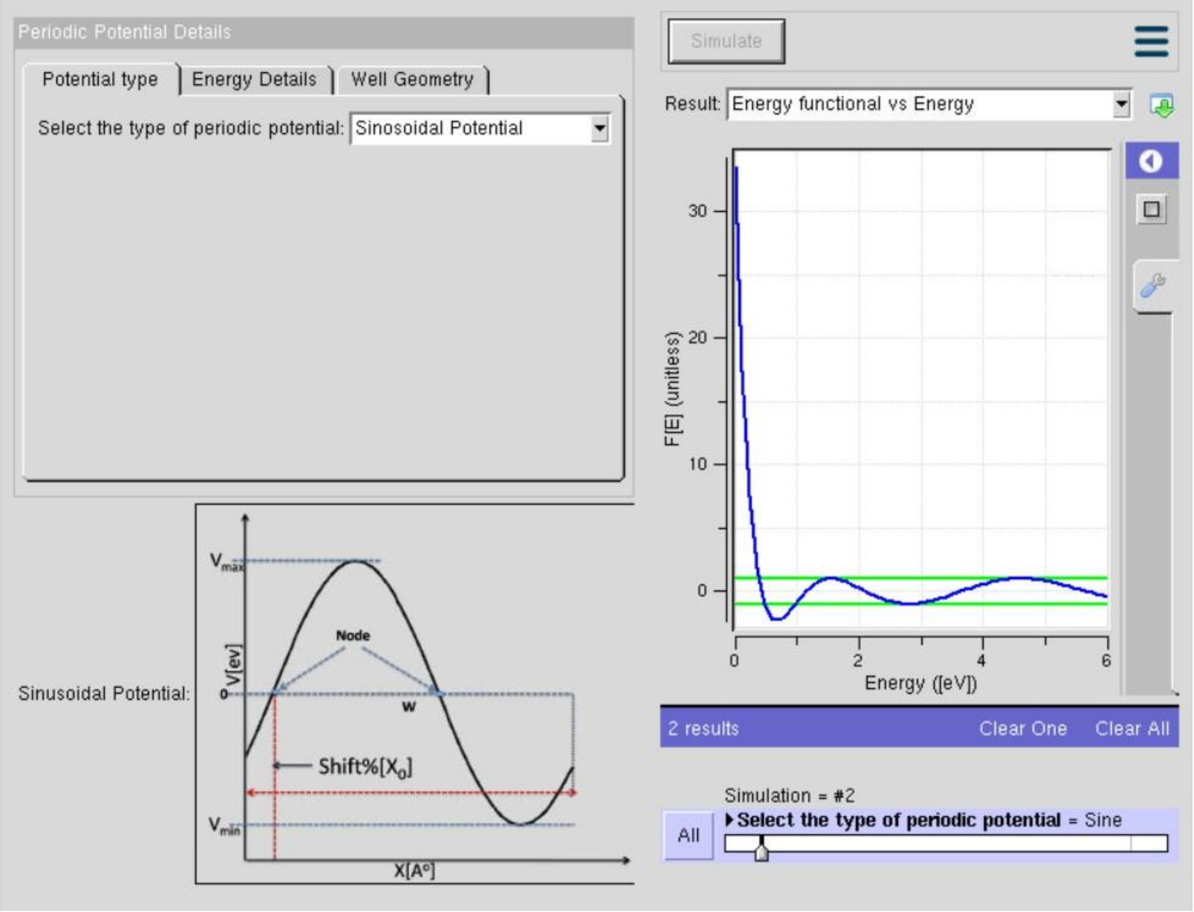
The graph illustrates the variation of the **Energy Functional (FE)** with respect to the **particle energy** in a periodic potential well. At low energy, the energy functional is very high and decreases rapidly as the particle energy increases. The curve then oscillates around the zero reference line, indicating changes in the system's stability. These oscillations represent the transition between **allowed and forbidden energy regions** in the periodic potential. This behavior demonstrates how the particle's energy is influenced by the periodic structure of the potential well.

Case Study 1: Simulate the output for Triangular well



The graph shows the variation of the **Energy Functional (FE)** with respect to **particle energy** for a **triangular periodic potential well**. At low energies, the energy functional is very high and decreases sharply as the energy increases. The FE curve then oscillates around the zero reference line, indicating changes in the stability of the particle. The intersections with the reference line represent the **allowed energy states (energy bands)** of the periodic structure. Overall, the graph demonstrates how the triangular potential influences the particle's energy distribution and band formation.

Case Study 2: Simulate the output for Sinusoidal Potential



The graph shows the variation of the **Energy Functional (FE)** with respect to **particle energy** for a **sinusoidal periodic potential**. At very low energies, the energy functional is high and decreases rapidly as the particle energy increases. The FE curve then oscillates around the zero reference line, indicating periodic changes in the particle's energy behavior. The points where the curve crosses the reference line correspond to **allowed energy states (energy bands)** in the sinusoidal potential. This graph illustrates how a sinusoidal periodic potential affects the energy distribution and stability of particles in the system.

In conclusion,The simulations of the **step, triangular, and sinusoidal periodic potentials** show that the energy functional decreases rapidly at low particle energies and then exhibits oscillatory behavior. In all three cases, the oscillations indicate the formation of **allowed and forbidden energy bands** due to the periodic nature of the potential. The particle becomes more stable at energy levels where the energy functional approaches zero. Although the potential shapes differ, they all produce similar band formation characteristics with slight variations in energy distribution. Overall, the results confirm that periodic potentials significantly influence particle stability and energy band formation in quantum systems.